Numerical Simulation of Ultrasonic Transducers in Aluminum Metal Casting

Author Name(s)
Department, Institution, City, Country

Corresponding author: email@domain.com

Highlights

- Coupled thermo–electro–mechanical workflow for casting-grade ultrasonic transducers.
- Preload- and contact-aware prestressed modal and harmonic response prediction.
- Electrical admittance outputs for calibration and validation against impedance measurements.
- Structural-to-acoustic mapping to molten aluminum for pressure-field screening and cavitation indices.
- Uncertainty and sensitivity guidance for tip temperature, preload, damping, and E(T).

Graphical Abstract (illustrative)

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Author Name(s) [to be added]

Affiliation(s), Department, Institution, City, Country

Corresponding author: email@domain.com

Abstract

Ultrasonic treatment of aluminum melts is increasingly used to promote degassing, grain refinement, and improved feeding by combining acoustic streaming with cavitation driven nucleation and inclusion breakup. Designing reliable high power ultrasonic transducers for casting environments is challenging because the transducer stack must remain within temperature limits while the sonotrode tip couples large amplitude vibration into molten aluminum and tooling. This paper presents a numerical simulation framework for predicting resonance behavior, vibration amplitude, thermal drift, and structural stress of ultrasonic transducers used in aluminum metal casting. The approach integrates (i) thermo mechanical analysis of the horn and stack under casting heat loads, (ii) prestressed modal analysis for resonance tracking, (iii) coupled piezoelectric harmonic response to estimate tip displacement and stress under voltage drive, and (iv) an acoustic field model in molten aluminum to assess pressure amplitude and process relevant indices such as cavitation threshold margin and streaming intensity proxies. A representative 20 kHz transducer–sonotrode system is used as a case study to demonstrate model setup, mesh convergence, boundary conditions, and parameter sensitivity. The results identify dominant design factors controlling hot resonance drift and elbow or fillet stress concentration, and provide practical simulation guidelines for robust transducer design and safe operation in aluminum casting lines.

Keywords: ultrasonic casting; aluminum melt; piezoelectric transducer; finite element method; thermo mechanical coupling; harmonic response; acoustic cavitation; resonance drift

Nomenclature

- f: resonance frequency (Hz)
- omega: angular frequency (rad/s)
- u: displacement vector (m)
- phi: electric potential (V)
- T: temperature (C)
- epsilon_th: thermal strain
- sigma_vm: von Mises stress (Pa)
- A_tip: tip displacement amplitude (m)
- p: acoustic pressure (Pa)
- rho: density (kg/m^3)
- c: sound speed (m/s)

1. Introduction

Ultrasonic melt treatment has shown strong potential to improve aluminum casting quality by reducing dissolved gases, breaking up oxide films and inclusions, and refining grain structure. These benefits are attributed to two coupled phenomena: (i) acoustic streaming, which increases melt convection and homogenization, and (ii) cavitation, which generates high local pressure gradients, micro jets, and nucleation sites.

Figure GA1. Representative workflow linking finite-element transducer response to melt acoustics. Replace with your own schematic or simulation figure for submission.

Abstract

Ultrasonic melt processing in aluminum casting requires stable, high-amplitude vibration delivery to molten metal while maintaining resonance under severe thermal gradients. Industrial transducer development

must control hot resonance drift, preload relaxation, contact nonlinearity at interfaces, and fatigue-critical stress hotspots at fillets and threaded joints. This paper presents a publication-grade numerical framework for simulating ultrasonic transducers used in aluminum metal casting. The framework couples steady-state/transient heat transfer, preload and contact-aware prestress, prestressed modal extraction, and fully coupled piezoelectric harmonic response to predict tip displacement amplitude, stress amplitude, and electrical admittance trends. The structural response is mapped to a melt acoustic model through velocity boundary conditions to estimate pressure amplitude fields and comparative cavitation/streaming indices. Reporting checklists for boundary conditions, mesh convergence, uncertainty quantification, and validation against impedance and laser vibrometry measurements are provided to support reproducible SCI submission and robust industrial design.

Keywords: ultrasonic aluminum casting; piezoelectric transducer; finite element analysis; prestress; frictional contact; thermo-mechanical coupling; harmonic response; admittance; melt acoustics; cavitation indicator; fatigue

Nomenclature

- u, U: displacement and complex displacement amplitude (m)
- f: frequency (Hz); omega: angular frequency (rad/s)
- T: temperature (C); T_ref: reference temperature (C)
- P0: bolt preload (N)
- sigma_vm: von Mises stress (Pa)
- A_tip: tip displacement amplitude (m)
- phi, V: electric potential and applied voltage (V)
- Y, Z: electrical admittance (S) and impedance (ohm)
- p: acoustic pressure (Pa); rho_f: melt density (kg/m^3); c_f: sound speed (m/s); k: wavenumber (1/m)

1. Introduction

Ultrasonic treatment of molten aluminum is applied to reduce dissolved gases, disperse oxide films and inclusions, and refine grain structure by leveraging acoustic streaming and cavitation-driven effects. In production casting lines, the transducer is typically mounted outside the hottest region while a metallic sonotrode transmits vibration to the melt or tooling.

High-power ultrasonic transducers are operated close to resonance to maximize sonotrode-tip amplitude. In aluminum casting, the horn is exposed to high temperature while the piezoelectric stack must remain within safe limits. Thermo–electro–mechanical coupling becomes dominant: temperature-dependent stiffness E(T) shifts resonance; thermal expansion modifies bolt preload; and interface contact conditions change damping and effective stiffness. Meanwhile, cyclic stresses near fillets and threads can drive high-cycle fatigue.

A credible simulation must therefore include: realistic thermal boundary conditions, preload and contact modeling, temperature-dependent properties, coupled piezoelectric actuation, and a validation plan using impedance/admittance and vibrometry. This manuscript consolidates these requirements into a reproducible workflow and provides journal-ready reporting templates.

1.1 Scope and novelty

- Coupled workflow (thermal → prestress/contact → prestressed modal → piezoelectric harmonic response → melt acoustics).
- Explicit outputs required for design decisions: f0, f_hot, Delta f, A_tip, sigma_vm hotspot stress, modal purity, and Y(f)/Z(f) trends.
- Uncertainty and sensitivity guidance for tip temperature, preload, friction coefficient, and damping.
- Validation plan and reporting checklists consistent with SCI expectations.

2. System Description and Modeling Assumptions

2.1 Transducer–horn assembly

A casting ultrasonic system generally includes a back mass, piezoelectric stack, front mass, and preload bolt, followed by a horn/sonotrode that couples vibration to molten aluminum. The model includes the full stack and horn up to the mounting boundary. The melt domain may be represented as an acoustic half-space or a bounded cavity representing a crucible or mold volume. Cooling and thermal barriers are represented through convection boundaries and (when applicable) material layers near the housing region.

For a frequency domain acoustic approximation in the melt, the pressure field p can be modeled by the Helmholtz equation:

$$\nabla^2 p + k^2 p = 0 \quad (5)$$

$$k = \omega / c \quad (6)$$

with density ρ and sound speed c of molten aluminum. A common coupling approach is to prescribe a normal velocity boundary condition at the sonotrode tip based on the structural solution: $v_n = j\omega U_n$. The local pressure amplitude $|p|$ is then used to evaluate process relevant indices. Cavitation onset depends on dissolved gas content and nuclei distribution; in simulation practice, a conservative threshold p_{th} may be used to compare relative performance (margin = $|p| / p_{th}$).

Figure 1. Illustrative casting integration photo/schematic (author-provided). Replace with your system photo, CAD, or simulation geometry.

2.2 Assumptions

- Metals: linear elasticity; report dynamic stress amplitude for fatigue assessment.
- Interfaces: bonded for ideal assembly; frictional contact (recommended) for preload-dependent stiffness and possible slip.
- Piezoelectricity: linear constitutive model with optional loss factors.

- Melt acoustics: linear Helmholtz model for comparative pressure fields; cavitation treated via indicator indices unless a full multiphase model is available.
- Material data: temperature dependence included for E(T) and thermal expansion; uncertainty bounds reported when data are incomplete.

3. Governing Equations

3.1 Heat transfer

(1) $\rho * c_p * dT/dt = \nabla \cdot (k_{th} \nabla T) + q_{dot}$

Boundary conditions may include prescribed temperature, convection $q = h*(T - T_{inf})$, and radiation $q = \varepsilon*\sigma_{SB}*(T^4 - T_{sur}^4)$. The computed $T(x,t)$ is transferred to the structural model through thermal strain and temperature-dependent material properties.

3.2 Prestress and structural dynamics

(2) $\varepsilon_{th} = \alpha(T) * (T - T_{ref})$

(3) $M * \ddot{u}(t) + C * \dot{u}(t) + K(T, P0) * u(t) = F(t)$

Prestress is solved first by applying bolt pretension P0 and the thermal strain field. The prestressed configuration is used for modal and harmonic analyses.

3.3 Harmonic response

(4) $[-\omega^2*M + j*\omega*C + K] * U = F(\omega)$

Damping may be represented by Rayleigh damping ($C = a*M + b*K$) or by a structural loss factor η. Parameters should be calibrated to match measured bandwidth in admittance curves.

3.4 Piezoelectric coupling and electrical response

(5) $S = s_E * T_\sigma + d^T * E$
(6) $D = d * T_\sigma + \varepsilon_T * E$

(7) $$\begin{bmatrix} K_{uu} & K_{u\phi} \\ K_{\phi u} & K_{\phi\phi} \end{bmatrix} \begin{bmatrix} u \\ \phi \end{bmatrix} = \begin{bmatrix} F \\ Q \end{bmatrix}$$

In the frequency domain, electrode current can be computed as $I = j*\omega*Q$ and admittance as $Y = I/V$. Admittance/impedance curves provide direct validation targets for ultrasonic transducer models.

3.5 Melt acoustics and coupling

(8) $\nabla^2 p + k^2 p = 0$
(9) $k = \omega / c_f$

(10) $v_n = j*\omega*U_n$ on the tip surface

(11) $M_{cav} = |p|_{max} / p_{th}$ (comparative cavitation margin)

4. Numerical Workflow and Reporting Requirements

1. Thermal: compute steady-state and/or transient temperature field for horn, masses, bolt, and housing.
2. Prestress/contact: apply P0 and T(x,t); obtain interface contact pressures; check separation/slip risk.
3. Prestressed modal: extract eigenmodes near f_target; compute modal purity eta_long at the tip.
4. Piezo harmonic: apply voltage; sweep frequency; extract A_tip(f), sigma_vm(f), and Y(f)/Z(f) trends.
5. Acoustics: map tip velocity boundary; compute |p| field and M_cav; optionally compute streaming proxy.
6. UQ/sensitivity: vary T_tip, P0, friction coefficient, E(T), and damping; report output bands.

4.1 Mesh convergence

Demonstrate convergence for both eigenfrequency and stress hotspot. Suggested criteria: $|\Delta f|/f < 0.1\%$ and peak sigma_vm change < 5% between mesh refinements. Use local refinement at fillets, thread roots, and contact interfaces.

4.2 Boundary condition checklist

- Thermal: specify T_tip or q_tip, convection coefficients (h) and ambient/coolant temperatures, radiation parameters if used.
- Mechanical: define mounting constraint type and region; justify if using fixed vs elastic support.
- Electrical: define electrode potentials and Vrms vs Vpk; state insulation boundaries.
- Interfaces: bonded vs contact, friction coefficient, contact algorithm, preload value and resulting contact pressure.
- Damping: Rayleigh or loss factor; calibration source (impedance bandwidth).

5. Results (Template for SCI-Style Reporting)

Replace bracketed values with your simulation outputs and add figures: temperature field, mode shape, stress map, admittance curve, and acoustic pressure contours.

5.1 Input parameters

Category	Parameter	Value	Notes
Operating	f_target	[ ] Hz	Generator target
Operating	Drive voltage	[ ] Vrms (or Vpk)	Define RMS/peak
Thermal	Tip boundary	T_tip = [ ] C or q_tip = [ ] W/m^2	Specify method
Thermal	Cooling boundary	h_cool = [ ] W/m^2K, T_cool = [ ] C	Housing/jacket
Assembly	Bolt preload	P0 = [ ] kN	From torque or stack spec
Interfaces	Contact model	bonded / frictional (mu = [ ])	State assumptions
Materials	Horn material	[ ]	Provide E(T), alpha(T), fatigue data
Melt	rho_f, c_f	[ ], [ ]	At casting temperature

5.2 Key outputs

Metric	Cold (25 C)	Hot (operating)	Notes
Resonance frequency	f0 = [ ] kHz	f_hot = [ ] kHz	Delta f = [ ] kHz ([ ] %)
Tip amplitude	A_tip = [ ] um	A_tip = [ ] um	At f_op and V
Peak von Mises stress	sigma = [ ] MPa	sigma = [ ] MPa	Hotspot location: [ ]
Modal purity	eta_long = [ ]	eta_long = [ ]	Axial/total at tip
Admittance (optional)	Y_peak = [ ]	Y_peak = [ ]	Compare to measurement
Acoustic pressure (optional)	|p|_max = [ ]	|p|_max = [ ]	M_cav = [ ] using p_th

6. Validation and Uncertainty Quantification

At minimum, validate (i) impedance/admittance curves near resonance at room temperature (and if feasible at elevated temperature), (ii) resonance drift under controlled heating, and (iii) tip amplitude measured by laser vibrometry at a specified voltage. Calibrate damping/loss factors and contact stiffness to match measured bandwidth and peak magnitudes without overfitting.

Dominant sensitivity factors typically include:

- T_tip and the shape of T(x): controls E(T) distribution and frequency drift.
- P0 and thermal relaxation: controls interface pressure and effective stiffness.
- Contact assumptions (bonded vs frictional, mu): can shift resonance and alter hotspot stress.
- Damping and loss factors: control amplitude and admittance bandwidth; must be reported and justified.

7. Discussion

Resonance management: tune cold resonance so that hot resonance aligns with f_target. If drift is large, adjust horn length or incorporate thermal barriers/cooling to reduce heat penetration.

Fatigue risk: report dynamic stress amplitude at hotspots. Increase fillet radii, smooth transitions, and locate threads away from displacement antinodes.

Amplitude delivery: ensure mode purity and stable interface pressure. Provide admittance curves as validation evidence and to support reported amplitude.

Melt coupling: pressure fields depend on tip velocity, immersion depth, and nearby boundaries; clearly state whether the acoustic model is linear and how coupling is applied.

8. Conclusions

A professional, reproducible numerical framework for ultrasonic transducers in aluminum metal casting has been presented. The workflow combines thermal analysis, preload/contact-aware prestress, prestressed modal extraction, coupled piezoelectric harmonic response, and melt acoustics mapping to predict resonance drift, amplitude delivery, admittance trends, and fatigue-relevant stresses. The included templates and checklists provide a direct path from simulation to SCI-style reporting.

References (insert your sources)

1 [1] Add references on ultrasonic melt treatment of aluminum (degassing, cavitation, grain refinement).
2 [2] Add references on high-power piezoelectric transducer design and impedance characterization.
3 [3] Add references on thermo-mechanical modeling of ultrasonic horns and resonance drift.
4 [4] Add references on acoustic modeling in molten metals and cavitation thresholds.